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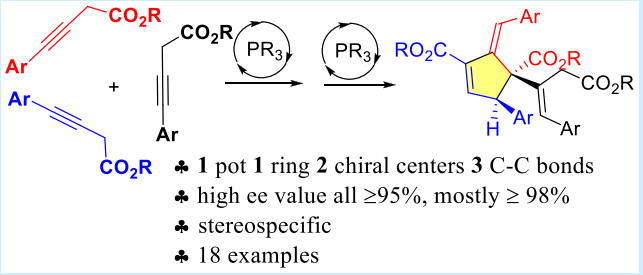
Enantioselective Phosphine-Catalyzed Trimerization of γ-Aryl-3-butynoates via Isomerization/[3 + 2] Cyclization/Michael Addition Cascade

Yujia Gao, Juan Zhang, Wenyu Shan, Weihong Fei, Jinzhong Yao, and Weijun Yao*

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ABSTRACT: We disclose an L-isoleucine-derived amide phosphine-catalyzed trimerization of γ-aryl-3-butynoates, which undergo an isomerization to allenoate, [3 + 2] cyclization, and Michael addition cascade. Exocyclopentene derivatives bearing an all-carbon quaternary stereocenter were constructed stereospecifically and enantioselectively. A wide variety of γ-aryl-3-butynoates could be employed to deliver optically pure cyclopentene derivatives in moderate to good yields with ee values of ≥95% and in most cases ≥98%.

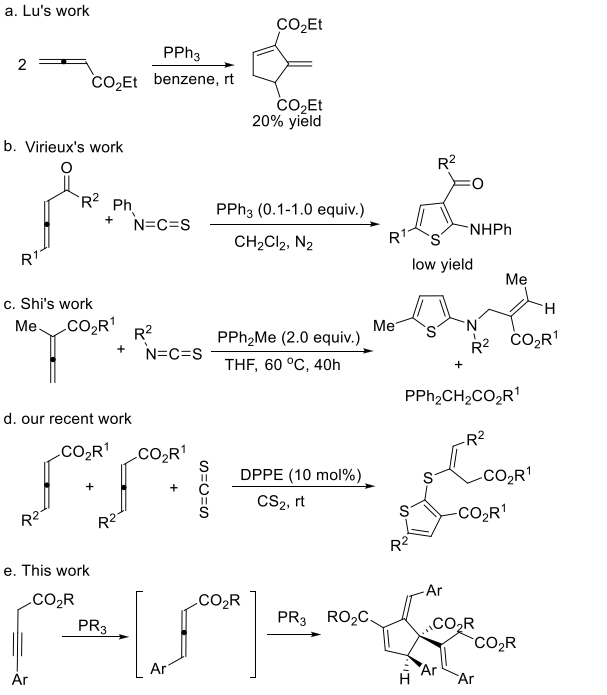


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The recent development of nucleophilic phosphine catalysis has witnessed its emergence as a powerful synthetic tool to access functionalized cyclic compounds.[1] Since Lu's pioneering work[2] on phosphine-catalyzed [3 + 2] cyclization of allenoates with electron-deficient olefins to construct cyclopentene derivatives in 1995, a wide variety of cyclizations involving allenoates and suitable dipolarophiles catalyzed by tertiary phosphines have been developed to deliver various carbocycles and heterocycles through [3 + 2],[3] [4 + 2],[4] and other annulations.[5] Compared with the wide exploration of electron-deficient olefins (C═C) and imines (C═N),[6] dipolarophiles containing cumulative double bonds, which could offer structural complexity of the cyclic products, have seldom been explored in this methodology. Lu's report[2] mentioned that self-cycloaddition of ethyl 2,3-butadienoate catalyzed by triphenylphosphine took place to provide the exocyclic compound in 20% yield (Scheme 1a). Similar results were also reported by Ma and coauthors in 2012.[7] Virieux[8] and Shi[9] reported phosphine-catalyzed [3 + 2] cycloadditions of electron-deficient allenes with isothiocyanates to synthesis 2-aminothiophenes (Scheme 1b,c). Recently, our group discovered a DPPE-catalyzed three-molecule two-component tandem reaction of allenoates and CS_2 to construct 2-thienyl vinyl sulfides through phosphine-catalyzed [3 + 2] cylyzation followed by Michael addition (Scheme 1d).[10] To the best of our knowledge, there are currently no reports on asymmetric phosphine-catalyzed transformations involving cumulative double bonds.

As part of our continuous effort to explore new reactions to construct cyclic skeletons,[11] we are interested in transformations of cumulative double bonds catalyzed by organocatalysis. Herein we report a stereospecific and enantioselective trimerization of γ-aryl-3-butynoates catalyzed by an L-

Scheme 1. Phosphine-Catalyzed Cyclizations Involving Cumulative Double Bonds



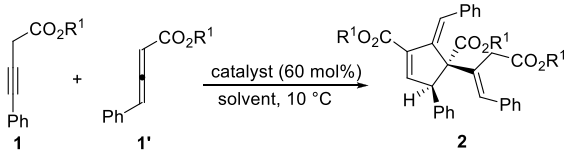
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Table 1. Optimization of the Reaction Conditions[a]



R^1 = Et **1a** : **1a'** = 20:1
R^1 = Me **1b** : **1b'** = 20:1

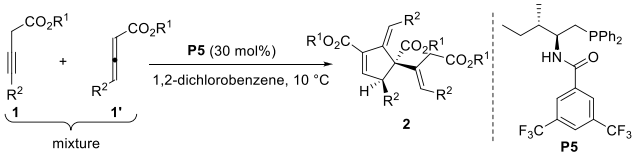
entry	R^1	catalyst	solvent	yield (%)[b]	ee (%)
1	Et	±**P1**	toluene	98	–
2	Et	**P1**	toluene	90	90
3	Et	**P2**	toluene	trace	–
4	Et	**P3**	toluene	trace	–
5	Et	**P4**	toluene	37	71
6	Et	**P5**	toluene	80	92
7	Et	**P6**	toluene	69	93
8	Et	**P5**	THF	43	75
9	Et	**P5**	DCM	48	89
10	Et	**P5**	$CHCl_3$	80	87
11	Et	**P5**	ethyl acetate	70	71
12	Et	**P5**	ether	75	56
13	Et	**P5**	DMF	trace	–
14	Et	**P5**	acetonitrile	trace	–
15	Et	**P5**	acetone	trace	–
16	Et	**P5**	mesitylene	75	92
17	Et	**P5**	xylenes	80	93
18	Et	**P5**	benzotrifluoride	76	95
19	Et	**P5**	chlorobenzene	74	94
20[c]	Et	**P5**	benzotrifluoride	64	93
21[c]	Et	**P5**	1,2-dichlorobenzene	72	97
22[c]	Me	**P5**	1,2-dichlorobenzene	85	98
23[c,d]	Me	**P5**	1,2-dichlorobenzene	85	98
24[c,e]	Me	**P5**	1,2-dichlorobenzene	81	98
25[c,f]	Me	**P5**	1,2-dichlorobenzene	88	94
26[g]	Me	**P5**	1,2-dichlorobenzene	70	97
27[h]	Me	**P5**	1,2-dichlorobenzene	46	97

[a]Unless otherwise stated, the reaction was carried out using a mixture of **1** and **1′** (0.2 mmol, 3.0 equiv), 60 mol % catalyst (0.04 mmol), and 1.5 mL of solvent at 10 °C overnight. [b]Isolated yields. [c]30 mol % catalyst was used. [d]0 °C. [e]−10 °C. [f]25 °C. [g]20 mol % catalyst was used. [h]10 mol % catalyst was used.

isoleucine-derived amide phosphine through an isomerization/[3 + 2] cyclization/Michael addition reaction cascade to construct exocyclopentene derivatives with an all-carbon quaternary stereocenter (Scheme 1e).

Considering the isomerization of γ-alkyl-substituted allenoates to dienes upon treatment with phosphine[12] and the low stability of γ-aryl-substituted allenoates, we chose γ-aryl-3-butynoates as the research substrates, which can isomerize to allenoates in the presence of phosphine.[13] We started our investigations by examining the reaction of ethyl 4-phenylbut-3-ynoate (**1a**), which is a mixture with ethyl 4-phenylbuta-2,3-dienoate (**1a′**), in the presence of common Lewis bases in toluene at 10 °C (ambient temperature). No reaction took place when PPh_3, DPPE, DABCO, or DMAP was involved. To our delight, the reaction proceeded smoothly to afford trimerization product **2a** in 98% yield when the bifunctional amide phosphine ±**P1** derived from DL-valine was employed (Table 1, entry 1). Encouraged by this result, we screened several different bifunctional phosphines based on L-valine, and the results showed that 3,5-bis(trifluoromethyl)benzamide-

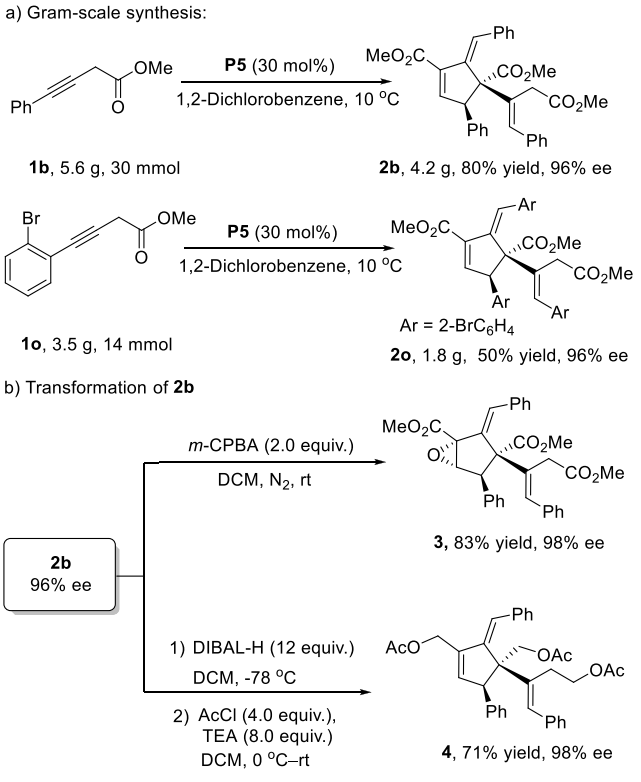
Table 2. Substrate Scope[a]



entry	**1** (R^2, R^1)	**1**:**1′**	yield of **2** (%)[b]	ee (%)
1	**1a** (Ph, Et)	20:1	72	98
2	**1b** (Ph, Me)	20:1	85	98
3	**1c** (Ph, Bn)	18:1	81	95
4	**1d** (4-MeC_6H_4, Me)	20:1	75	>99
5	**1e** (4-EtC_6H_4, Me)	8:1	84	98
6	**1f** (4-MeOC_6H_4, Me)	20:1	81	95
7	**1g** (4-FC_6H_4, Me)	7:1	71	>99
8	**1h** (4-ClC_6H_4, Me)	1.66:1	75	>99
9	**1i** (4-BrC_6H_4, Me)	11:1	73	99
10	**1j** (4-$CF_3C_6H_4$, Me)	1:1.25	68	99
11	**1k** (3-MeC_6H_4, Me)	5:1	76	99
12	**1l** (3-FC_6H_4, Me)	8:1	78	>99
13	**1m** (3-ClC_6H_4, Me)	13:1	75	>99
14	**1n** (2-ClC_6H_4, Me)	3.3:1	25	98
15	**1o** (2-BrC_6H_4, Me)	12:1	54	96
16	**1p** (piperonyl, Me)	>20:1	77	98
17	**1q** (2-naphthyl, Me)	19:1	80	99
18	**1r** (3-thiophenyl, Me)	20:1	83	98
19	**1s** (*n*Bu, Me)	>20:1	–	–
20	**1t** (*t*Bu, Me)	8:1	–	–

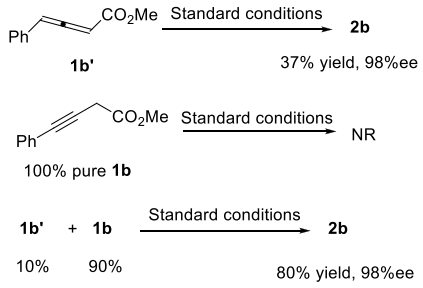
[a]Unless otherwise stated, the reaction was carried out using mixture of **1** and **1′** (0.4 mmol, 3.0 equiv) and 30 mol % **P5** (0.04 mmol) in 3 mL of 1,2-dichlorobenzene at 10 °C overnight. [b]Isolated yields.

Scheme 2. Gram-Scale Synthesis of 2 and Transformation of 2b



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Scheme 3. Control Experiments

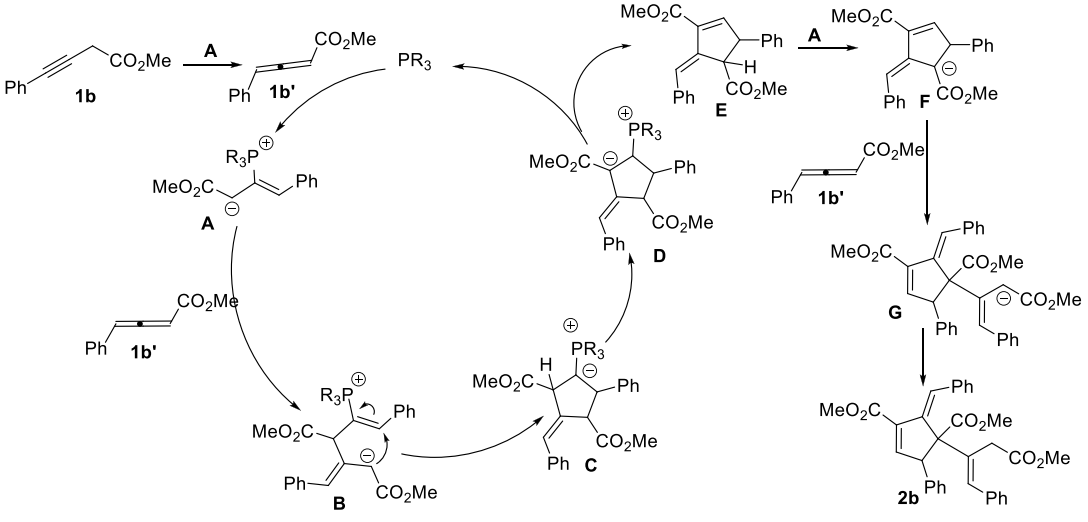


modified phosphine **P1** gave the best results, affording the product in 90% yield with 90% ee (Table 1, entry 2), compared with phosphines substituted with other functional groups such as sulfonamide, thiourea, and pivalamide (Table 1, entries 3−5). Moreover, 3,5-bis(trifluoromethyl)benzamide-modified phosphines **P5** and **P6** derived from L-isoleucine and L-threonine, respectively, were tested, and both worked well in this process to give the product **2a** in 80% and 69% yield, respectively, with high enantioselectivity (Table 1, entries 6 and 7). Taking into account the synthesis efficiency of the phosphine catalyst, we chose **P1** and **P5** as potential catalysts to investigate the effect of the solvent and temperature and to improve the yield and enantioselectivity in this novel trimerization process (see the SI for details). The catalyst **P5** derived from L-isoleucine performed better than **P1** in each issue of various parallel comparisons. Solvent screening revealed that aromatic solvents gave the highest enantioselectivity (up to 97% ee; Table 1, entries 16−21) and solvents with moderate or weak polarity led to moderate to good enantioselectivity (Table 1, entries 8−12) while strongly polar solvents (DMF, acetonitrile, and acetone) restrained this reaction (Table 1, entries 13−15). When methyl 4-phenylbut-3-ynoate (**1b**) was used instead, the target product **2b** was isolated in 85% yield with 98% ee in 1,2- dichlorobenzene catalyzed by **P5** (Table 1, entry 22). Using a lower temperature did not increase the enantioselectivity (Table 1, entries 22−24), while using a higher temperature decreased the enantioselectivity slightly to 94% ee, although the yield was improved slightly to 88%. Furthermore, when the catalyst loading was decreased to 20 mol %, the yield of the target product **2b** declined to 70% with the ee retained (Table 1, entry 26); when the catalyst loading was decreased to 10 mol %, the yield dropped to 46%, and the 3-butynoate was not consumed completely even with a prolonged reaction time (Table 1, entry 27). Finally, the optimal conditions were established for this reaction as 30 mol % **P5** as the catalyst and 1,2- dichlorobenzene as the solvent at 10 °C.

Having identified the optimal conditions, we next explored the generality of the substrate scope of γ-aryl-3-alkynoates. As shown in Table 2, it was found that different ester groups, such as ethyl (**1a**), methyl (**1b**), and benzyl (**1c**) were all compatible with this reaction, delivering the desired products in 72−85% yield with up to 98% ee. Moreover, substrates with either an electron-withdrawing or electron-donating group on the phenyl ring underwent this phosphine-catalyzed tandem reaction smoothly to furnish the desired products **2d−k** and **2p** in 68−84% yield with 95−99% ee. However, aromatic rings containing a substituent at the *ortho* position gave lower yields (**2n** and **2o**) because of the steric hindrance. The reaction was also applicable to 2-naphthyl- and 3-thiophenyl-containing substrates, providing the corresponding cycloadducts **2q** and **2r** in good yields with extremely high enantioselectivity. Unfortunately, when γ-butyl alkynoate **1s** was employed in this reaction, a complicated mixture was obtained, and HRMS showed no trimerization product. No reaction took place when γ-*tert*-butyl alkynoate **1t** was involved. Notably, the stereospecificity of this reaction was evident, as only one stereoisomer of cyclopentene derivative **2** was detected in all of the examples despite the presence of two continuous chiral centers with an all-carbon quaternary stereocenter. The absolute configuration of product **2a** was determined by single-crystal X-ray diffraction.

Further evaluation of the catalytic efficiency indicated that this phosphine-catalyzed reaction could be easily enlarged to the gram scale. **2b** and **2o** were obtained in 4.2 and 1.8 g, respectively, under the standard conditions with the yield and enantioselectivity maintained (Scheme 2a). Regioselective epoxidation of **2b** was realized in the presence of *m*-CPBA to provide product **3**[14] in 83% yield with 98% ee (Scheme 2b). Interestingly, product **3** could also be obtained by ozonolysis in

Scheme 4. Proposed Mechanism



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30% yield. Furthermore, **2b** could be reduced to the alcohol by DIBAL-H at −78 °C, and subsequent esterification with acetyl chloride furnished compound **4** in 71% yield with 98% ee.

To understand this phosphine-catalyzed cascade reaction better, control experiments were carried out (Scheme 3). Allenoate **1b′** was employed in this tandem process to give the product **2b** in 37% yield with 98% ee, while absolutely pure alkynoate **1b** did not work at all under the standard conditions. When a catalytic amount of **1b′** was added to alkynoate **1b**, the reaction took place smoothly as normal. This revealed that the phosphine could not react with the 3-alkynoate and could not isomerize the 3-alkynoate to the allenoate.

On the basis of previously reported results and our experiments, a plausible reaction mechanism is depicted in Scheme 4. Initially, nucleophilic phosphine attacks the trace amount of allenoate **1b′** to form zwitterionic intermediate **A**, which plays two important roles: it acts not only as a Bronsted base to isomerize the alkynoate to the allenoate[15] but also as a key intermediate to take part in the cyclization. Zwitterionic intermediate **A** reacts with another allenoate to yield intermediate **B**, followed by an intramolecular Michael addition to give intermediate **C**. After a 1,2-proton shift to give **D**, the catalyst is recycled, and simultaneously, cyclization adduct **E** is released and then deprotonated by zwitterionic intermediate **A** in the system to generate carbanion **F**. Finally, Michael addition of **F** with the third allenoate provides the final product **2b**.

In conclusion, we have developed an L-isoleucine-derived amide phosphine-catalyzed tandem reaction of γ-aryl-3-butynoates, which undergo isomerization to the corresponding allenoates followed by [3 + 2] cyclization and Michael addition. Cyclopentene derivatives with an exocyclic double bond and bearing an all-carbon quaternary stereocenter were constructed stereospecifically and enantioselectively. A wide variety of γ-aryl-3-butynoates could be employed in this process to deliver almost optically pure cyclopentene derivatives in moderate to good yields.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.1c02197.

Experimental procedures, characterization data, crystallographic data for **2a** and **3**, and copies of ^{1}H and ^{13}C NMR spectra for all new compounds (PDF)

Accession Codes

CCDC 2082657 and 2082658 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, U.K.; fax: +44 1223 336033.

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Notes

The authors declare no competing financial interest.

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